

Senior Design Project – UAV Powertrain Health Monitoring (Fall 2025)

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UAV-HealthMonitoring

UAV Health Monitoring

Team Members

- Prishha Chawla (CS) – chawlaps@uc.edu
- Siddharth Urankar (CS) – urankasj@mail.uc.edu
- Ally Blair (ME) – blairar@mail.uc.edu

Faculty/Industry Advisors

- Dr. Manish Kumar (UC, Mechanical Engineering)
- Dr. Manuel Arias Chao (ZHAW, Switzerland)
- Dr. Chetan Kulkarni (NASA Ames)

Project Summary

Our project develops a UAV powertrain health monitoring framework using machine learning to detect motor anomalies and predict battery degradation.

- Motor health: semi-supervised 1D CNN Autoencoder trained on healthy BLDC motor acoustics.
- Battery health: supervised DeepONet trained on voltage–current profiles to estimate degradation, Remaining Useful Life (RUL), and remaining flight time.

This system will integrate both models to provide real-time in-flight diagnostics and preflight readiness checks, supporting UAV reliability in mission-critical operations.

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